

Generation Interconnection Request System Impact Study Report

For Queue Project Q23-2348

Windmill 100 kV
August 25, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2348. The Interconnection Customer proposes a 179.3 MW solar photovoltaic facility interconnecting at the Windmill 100 kV substation in the Ohio area. The request was received on September 10, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 376 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,113,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: W Junction-East Broad Street 100.0 kV
Contingency: John C. Eisinger-Isabella 100.0 kV L
Loading: 123.5% (Rating: 158 MVA)
DFAX: 7.4%
Pre-project loading: 115.1%
2. Facility: Jug St-West Jct 100.0 kV
Contingency: John C. Eisinger Substation-Isabella 100.0 kV
Loading: 122.8% (Rating: 132 MVA)
DFAX: 7.4%
Pre-project loading: 112.7%
3. Facility: E Broad Street Substation-Windmill 100.0 kV
Contingency: Norwich-Barrett Substation 500.0 kV
Loading: 107.4% (Rating: 162 MVA)
DFAX: 58.4%
Pre-project loading: 42.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Junction-East Broad Street 100.0 kV	Rebuild line with higher-capacity conductor	\$9,232,000	100% (\$9,232,000)
Jug St-West Jct 100.0 kV	Rebuild line with higher-capacity conductor	\$4,333,000	100% (\$4,333,000)
E Broad Street Substation-Windmill 100.0 kV	Reconductor line	\$601,000	91% (\$548,000)
	Total allocated to project		\$14,113,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.